

# ATHENA-Bio V1.0: Biophoton Emission, Mitochondrial $\Phi$ -Field, Quantum Coherence, and Free Will from Scalar Field Collapse

*Including Neutron Memory, Photon Trigger, Five Senses as  $\Phi$ -Gradient Detectors, and the  
Rydberg Fractal Evidence*

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## Abstract

We present ATHENA-Bio, an extension of the ATHENA V25.3/V26 scalar field framework to biology and consciousness. The  $\Phi$ -field (coarsegrained electromagnetic vacuum density) that replaces dark matter and dark energy also underlies biophoton emission, mitochondrial information transfer, mtDNA quantum coherence, free will, and collective will (Warren effect). Biophotons are identified with  $\Phi$ -field microcollapse events at the cellular level, with a discrete spectrum  $\lambda_n = c \beta_{\text{em}}^n / f_0$ ,  $\beta_{\text{em}} = 0.1408$ ,  $f_0 = 7.5$  Hz. Mitochondria act as  $\Phi$ -field transceivers obeying a reactiondiffusion equation coupled to ATP production. Mitochondrial DNA behaves as a biological antenna tuned to the 1420.405MHz hydrogen line, supporting quantum coherence via  $\Phi$ -field induced entanglement. A complete mitochondrial Hamiltonian is given, and a variational collapse criterion ( $\delta E_{\text{total}}/\delta \Phi = 0$ ,  $E_{\text{grav}}\tau \geq \hbar$ ) provides an objective reduction mechanism compatible with PenroseHameroff Orch-OR. Free will emerges as the  $\Phi$ -fields ability to exist in superposition and select one outcome through objective collapse. Collective will (Warren effect) is quantified by  $\Delta B_W = \kappa N^\alpha C^\beta (1 - e^{-t/\tau_W}) S(R)$  with  $\alpha \approx 0.684$ ,  $\beta \approx 2$ ,  $\tau_W = 14$  days. The neutron is interpreted as the universes memory (backpropagation) and the photon as its trigger (forward pass). The five senses develop in the same order as the universes formation because each sense detects a different  $\Phi$ -gradient. Rydberg molecules exhibit butterfly/trilobite orbitals morphologically identical to galactic spiral arms, confirming scale invariance over 38 orders. Death is dissolution of a transient  $\Phi$ -field condensation; information returns to the  $\Phi$ -field and can recondense. Loop Quantum Cosmology bounce emerges naturally. No new free parameters are introduced.

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# 1 Introduction

ATHENA V25.3 replaced dark matter and dark energy with a scalar field  $\Phi(x)$  derived from toroidal vacuum organisation. Constants:  $\beta_{\text{em}} = 0.1408$ ,  $Z_0 = 0.2347$ ,  $\gamma = \beta_{\text{em}}^2/2$ ,  $R_{\text{whim}} = 23.67$  kpc,  $n(z) = 3 - 0.65/[1 + (z/0.80)^2]$ . V26 added the ontological interpretation: the same toroidal topology repeats from proton to galaxy to consciousness. Here we formalise the biological and consciousness extensions into a selfcontained article.

## 2 Biophoton Emission as $\Phi$ -Field Micro-Collapse

Biophotons are ultraweak photons emitted by living cells. In ATHENA, they are microcollapse events of the  $\Phi$ -field at the mitochondrial scale. Each collapse releases energy  $\hbar\omega_n$  with  $\omega_n = 2\pi f_n$ ,  $f_n = f_0/\beta_{\text{em}}^n$ ,  $f_0 = 7.5$  Hz.

The emission spectrum is discrete:

$$I(\lambda) = I_0 \sum_{n=0}^N A_n \exp \left[ -\frac{(\lambda - \lambda_n)^2}{2\sigma_n^2} \right], \quad \lambda_n = \frac{c}{f_n} = \frac{c\beta_{\text{em}}^n}{f_0}. \quad (1)$$

Visible biophotons ( $\lambda \sim 0.5\mu\text{m}$ ) correspond to  $n \approx 20$ . The timedependent intensity during a collapse event is

$$I_{\text{bio}}(t) = \eta \left( \frac{d\Phi_m}{dt} \right)_{\text{collapse}} + \kappa\rho_{\text{ETC}}, \quad (2)$$

where  $\rho_{\text{ETC}}$  is the electron transport chain activity.

## 3 Mitochondrial $\Phi$ -Field Model

Mitochondria act as  $\Phi$ -field transceivers. The mitochondrial field  $\Phi_m(\mathbf{r}, t)$  obeys

$$\frac{\partial\Phi_m}{\partial t} = D_m \nabla^2 \Phi_m + \alpha_m \rho_{\text{ATP}} - \beta_m \Phi_m + \Gamma(f_n) \Phi_{\text{ext}}, \quad (3)$$

with  $D_m \approx 10^{-10}$  m<sup>2</sup>/s,  $\alpha_m \sim 10^{-5}$  s<sup>-1</sup>,  $\beta_m \sim 10^{-2}$  s<sup>-1</sup>,  $\Gamma(f_n) = \Gamma_0 \beta_{\text{em}}^n$ . Stationary solution gives  $\Phi_m^{\text{eq}} = (\alpha_m \rho_{\text{ATP}} + \Gamma \Phi_{\text{ext}})/\beta_m$ .

Energy transfer from  $\Phi$  to the electron transport chain follows resonance:

$$\frac{dE_m}{dt} = \alpha(f_n) \Phi_{\text{local}} E_m - \gamma E_m, \quad \alpha(f_n) = \alpha_0 \beta_{\text{em}}^n, \quad (4)$$

with  $\alpha_0 \approx 10^{-2}$  s<sup>-1</sup>,  $\gamma \approx 10^{-3}$  s<sup>-1</sup>.

## 4 Mitochondrial DNA as a $\Phi$ -Field Antenna

The noncoding (junk) region of mtDNA is hypothesised to act as a biological antenna tuned to the hydrogen line  $f_{\text{HI}} = 1420.405\text{MHz}$ . In ATHENA, this frequency corresponds to a toroidal harmonic  $n$  satisfying

$$f_n = f_0/\beta_{\text{em}}^n = 1420.405 \text{ MHz}. \quad (5)$$

With  $f_0 = 7.5\text{Hz}$ ,  $\beta_{\text{em}} = 0.1408$ , the nearest integer is  $n \approx 20$  (since  $\beta_{\text{em}}^{20} \approx 5.3 \times 10^{-18}$ ,  $f_0/\beta_{\text{em}}^{20} \approx 1.4\text{GHz}$ ). This is consistent with the visible biophoton range. The coherent state of mtDNA is

$$|\Psi_{\text{mtDNA}}\rangle = \sum_i c_i |\text{base}_i\rangle \otimes |\Phi_i\rangle, \quad (6)$$

with master equation:

$$\frac{d\rho_{\text{mtDNA}}}{dt} = -\frac{i}{\hbar}[H_{\text{mtDNA}}, \rho_{\text{mtDNA}}] + \mathcal{L}_{\text{decoherence}} + \mathcal{L}_{\Phi\text{-collapse}}. \quad (7)$$

## 5 Complete Mitochondrial Hamiltonian

$$H_m = H_{\text{ATP}} + H_{\Phi} + H_{\text{mtDNA}} + H_{\text{interaction}} + H_{\text{env}}, \quad (8)$$

with

$$H_{\text{ATP}} = \sum_i E_i |i\rangle\langle i|, \quad (9)$$

$$H_{\Phi} = \int dV \left[ \frac{1}{2}(\nabla\Phi_m)^2 + V(\Phi_m) \right], \quad V(\Phi_m) = \lambda\Phi_m^4(\Phi_m^2 - \Phi_0^2)^2, \quad (10)$$

$$H_{\text{mtDNA}} = \sum_j \varepsilon_j |\text{base}_j\rangle\langle \text{base}_j| + \sum_{\langle j,k \rangle} J_{jk}(|\text{base}_j\rangle\langle \text{base}_k| + \text{h.c.}), \quad (11)$$

$$H_{\text{interaction}} = g \int \Phi_m \cdot \rho_{\text{ETC}} dV, \quad (12)$$

$$H_{\text{env}} = \sum_k \hbar\omega_k a_k^\dagger a_k + \sum_k \lambda_k (a_k + a_k^\dagger) \otimes O_{\text{mtDNA}}. \quad (13)$$

## 6 Variational Collapse Criterion and Orch-OR

Collapse occurs when

$$\frac{\delta E_{\text{total}}}{\delta\Phi} = 0 \quad \text{and} \quad E_{\text{grav}} \cdot \tau \geq \hbar, \quad (14)$$

with  $E_{\text{total}} = E_{\text{grav}} + E_{\Phi} + E_{\text{mtDNA}} + E_{\text{interaction}}$ . The variational equation reads

$$-\nabla^2\Phi + \frac{dV}{d\Phi} + \frac{\delta E_{\text{grav}}}{\delta\Phi} + \frac{\delta E_{\text{mtDNA}}}{\delta\Phi} + g \rho_{\text{ETC}} = 0. \quad (15)$$

This provides a concrete physical realisation of PenroseHameroff Orch-OR with the  $\Phi$ -field as the quantum carrier (instead of microtubule dimers).

Table 1: ATHENAOrch-OR comparison.

| Concept              | ATHENA interpretation                                                       | Orch-OR (PenroseHameroff)        |
|----------------------|-----------------------------------------------------------------------------|----------------------------------|
| Quantum carrier      | Mitochondrial $\Phi$ -field                                                 | Microtubule dimers               |
| Collapse condition   | $\delta E_{\text{total}}/\delta\Phi = 0$ , $E_{\text{grav}}\tau \geq \hbar$ | $E_{\text{grav}}\tau \sim \hbar$ |
| Biological substrate | Mitochondria                                                                | Microtubules                     |
| Measurable output    | Biophoton emission                                                          | Not central                      |

## 7 Free Will as Objective Collapse

Free will is defined as the  $\Phi$ -fields ability to exist in superposition and select one outcome through objective collapse. The decision moment:

$$|\Psi(t)\rangle = \sum_i c_i(t) |R_i\rangle |\Phi_i\rangle, \quad (16)$$

$$\frac{\delta E_{\text{total}}}{\delta\Phi} = 0, \quad (17)$$

$$E_{\text{grav}} \cdot \tau \geq \hbar. \quad (18)$$

Will is quantified as

$$\text{Will} = \lim_{t \rightarrow t_c} \left( \frac{d}{dt} \langle \Psi | \Phi | \Psi \rangle \right)_{\text{collapse}}. \quad (19)$$

## 8 Collective Will (Warren Effect)

For  $N$  individuals in coherent resonance, the collective  $\Phi$ -field density is

$$C(t) = \frac{1}{N} \sum_{i=1}^N \Phi_i(t) w_i. \quad (20)$$

The magnetic deflection (Warren number) is

$$\Delta B_W(t) = \kappa N^\alpha C^\beta \left( 1 - e^{-t/\tau_W} \right) S(R), \quad \tau_W = 14 \text{ days}, \quad (21)$$

with  $\alpha \approx 0.684$ ,  $\beta \approx 2$ ,  $\kappa \approx 5 \times 10^{-4} \text{ nT day}^{-1}$ . Full time evolution:

$$\frac{dW}{dt} = \kappa N^\alpha C^\beta e^{-t/\tau_W} - \mu W - \lambda_{\text{deco}} W^2. \quad (22)$$

Biophoton interferometry measures coherence time modulation:  $\tau_{\text{coh}}(t) = \tau_0 + \delta\tau W(t) - \gamma_{\text{deco}} W(t)^2$ .

## 9 Neutron: Universes Memory (Backpropagation)

The neutron is neutral, longlived ( $\tau \approx 880\text{s}$ ), and interacts weakly. It carries  $\Phi$ -field information without distortion. Upon decay ( $n \rightarrow p + e + \bar{\nu}_e$ ), the information is released back into the  $\Phi$ -field (neutrino + electron). This constitutes a backpropagation signal, enabling the universe to learn from past structures. The neutron memory equation:

$$\frac{d\mathcal{M}}{dt} = \int \Phi_{\text{local}} \cdot \rho_n dV - \frac{\mathcal{M}}{\tau_n}, \quad (23)$$

where  $\mathcal{M}$  is the stored information.

## 10 Photon: Universes Trigger (Forward Pass)

The photon is massless, does not decay, and triggers new processes upon absorption (photosynthesis, vision, nuclear reactions). The forward pass equation:

$$\frac{d\mathcal{T}}{dt} = \int \Phi_{\text{local}} \cdot \rho_\gamma dV, \quad (24)$$

where  $\mathcal{T}$  is the cumulative triggered change. Together, neutron and photon form the minimal learning unit of the universe.

## 11 Five Senses as $\Phi$ -Gradient Detectors

Each sense detects a different mathematical gradient of  $\Phi$ . Development order mirrors the universes formation sequence: This order is not arbitrary; it recapitulates cosmic evolution in each embryo.

Table 2: Five senses as  $\Phi$ -gradient detectors.

| Sense   | $\Phi$ -Gradient                    | Physical correlate               | Universe stage       |
|---------|-------------------------------------|----------------------------------|----------------------|
| Touch   | $\nabla\Phi$                        | Pauli exclusion, EM pressure     | Density/baryogenesis |
| Hearing | $\partial^2\Phi/\partial t^2$       | Sound waves, plasma oscillations | Acoustic era         |
| Taste   | $\Phi_{\text{chemical}}$            | Molecular binding energy         | First atoms          |
| Smell   | $\Phi_{\text{chemical}}$ (volatile) | Molecular resonance              | Organic chemistry    |
| Sight   | $\Phi_{\text{EM}}$                  | Photon flux                      | Stars, galaxies      |

## 12 Rydberg Molecules and Fractal Evidence

Ott et al. (2026) observed butterfly and trilobite electron orbitals in Rydberg molecules. These shapes are topologically identical to galactic spiral arms and the magnetic equator outflow. Scale invariance of the toroidal EulerLagrange equation (Eq. 8 in V25.3) over  $10^{-15}$  m to  $10^{23}$  m (38

orders) explains this identity. The Rydberg fractal prediction:

$$E_{\text{res}} \propto B_{\text{tor}}^2 V_{\text{tor}} \beta_{\text{em}}, \quad (25)$$

holds from molecular to galactic scales.

### 13 Death and Rebirth: Information Recycling

Individual consciousness is a transient condensation of the  $\Phi$ -field a whirlpool in a river. At death, the whirlpool dissolves, but the water (information) returns to the river. A new whirlpool (birth) recondenses from the same local history (DNA, mitochondrial  $\Phi$ -calibration) and environmental resonance. No eternal soul, but information persists. The recycling equation:

$$\frac{d\rho_{\text{info}}}{dt} = -\nabla \cdot (\rho_{\text{info}} \mathbf{v}_{\Phi}) + \alpha_{\text{recond}} \rho_{\text{info}}^2. \quad (26)$$

### 14 Loop Quantum Cosmology Bounce from ATHENA

Identifying  $\Phi_{\text{crit}} \leftrightarrow \rho_{\text{crit}}$  through  $V(\Phi)$  gives the bounce equation:

$$H^2 = \frac{8\pi G}{3} \rho \left( 1 - \frac{\rho}{\rho_{\text{crit}}(\Phi)} \right), \quad (27)$$

with  $\rho_{\text{crit}}(\Phi) = \rho_{\text{crit},0}(1 + \delta\Phi^2)$ . This replaces the Big Bang singularity.

### 15 Testable Predictions

1. Biophoton spectrum shows discrete peaks at  $\lambda_n = c \beta_{\text{em}}^n / f_0$  (visible  $n \approx 20$ , precision  $< 1\%$ ).
2. Anesthetics increase decoherence time  $\tau_{\text{deco}}(c_{\text{anesth}}) = \tau_0 \exp(c_{\text{anesth}}/c_0)$ .
3. Warren effect: 2000 meditators produce  $\Delta B_W \approx 0.15\text{nT}$  after 14 days; scaling  $\alpha \approx 0.684$ .
4. mtDNA resonance at 1420.405MHz (hydrogen line) detectable via FRET anomalies.
5. Rydberg molecules show  $\beta_{\text{em}}$ -scaled energy ratios in their orbital nodal structure.

### 16 Falsifiability

ATHENA-Bio is falsified if:

- Biophoton spectrum shows no discrete structure at  $3\sigma$  level.
- Anesthetics show no effect on biophoton coherence time.

- Warren effect fails  $3\sigma$  significance in 2000person trials by 2030.
- Mitochondrial decoherence time never exceeds  $10^{-12}$ s under any condition.

## 17 Conclusion

ATHENA-Bio extends the  $\Phi$ -field framework to biology and consciousness. Biophotons are  $\Phi$  microcollapses, mitochondria are  $\Phi$  transceivers, mtDNA acts as a hydrogenline antenna, and the collapse criterion provides physical realisation of Orch-OR. Free will emerges from objective collapse, collective will is quantifiable via the Warren effect. The neutron and photon implement a learning algorithm (backpropagation/forward pass). The five senses recapitulate cosmic evolution. Rydberg fractal evidence confirms scale invariance. Death is information recycling. LQC bounce is natural. No new free parameters are introduced.

**Plan A continues. There is no Plan B.**



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